

Patrick Stock

DSP Software Engineer

pstockdev@gmail.com | pstock.dev | github.com/pstock7 | linkedin.com/in/patricketstock

DSP software engineer focused on radio signal processing, containerized services, and software-defined radio integration. Experienced developing and testing mission-critical applications and supporting on-site integration events to ensure systems are ready for deployment.

Skills

Languages: Python, C++, C, Rust, Lua, JavaScript, Java, Nix, Bash

Signal Processing: X-Midas, signal detection and validation, spectrum analysis, software-defined radio integration (Ettus, HTL)

Systems & Messaging: Linux, Docker, containerized distributed services, ActiveMQ

Experience

DSP Software Engineer

June 2024–Present

Parsons — Herndon, Virginia

- Serve as a software subject-matter expert at on-site integration events, resolving SDR configuration, signal detection, and distributed-service issues across containerized systems.
- Develop Python services for processing radio IQ data and coordinating Docker containers through an ActiveMQ message broker.
- Designed and implemented an end-to-end RF test service that transmits known signals between system instances and validates resulting detections through the message broker.
- Act as the primary software point of contact for a customer project, supporting integration questions and maintaining continuity for its software architecture.

DSP Software Intern

Summer 2022 & Summer 2023

Parsons — Herndon, Virginia

- Developed a containerized Python application for real-time spectrum visualization, radio control, and signal recording and analysis, including waterfall displays, time-selective and frequency-selective recording, storage-capacity estimates, and tools for reviewing completed recordings.
- Operated SDR equipment and served as a hardware and software resource during a customer demonstration.
- Created a VPN service enabling radios to communicate across networks.

Education

Master of Science in Computer Engineering

August 2024–Present

Virginia Tech — Online

Focus: Networking & Cybersecurity

Courses Completed: Information Theory; Network Architecture and Protocols; System and Software Security

Bachelor of Science in Computer Engineering

August 2020–May 2024

Virginia Tech — Blacksburg, Virginia

Focus: Machine Learning | **Minors:** Computer Science, Cybersecurity, Math

Courses Completed: Network Application Design; Advanced Machine Learning; Computer Organization and Architecture; Computer and Network Security; Embedded Systems; Computer Vision; Cryptography; Artificial Intelligence and Engineering Applications